Resonance

Example 11

For the RLC series with R = 12 ohm, L = 0.15H, and C =22 uF, determine the impedance at i) resonance frequency, ii) 10 Hz below resonant frequency and iii) above resonant frequency.

fr = 87.6119 Hz, Zr = 12 ohm, Z_{fr-10} = 23.3786<-59.1, Z_{fr+10} = 21.5369<56.1 .

Example 12

For the RLC series with R = 5 ohm, L = 0.03 H, and C = 100 uF, determine the frequency at which the circuit resonates. Also find the quality factor, voltage across inductance and voltage across capacitance at resonance

fr = 91.8882 Hz, Qr = 3.4641, Vl = Vc = 69.282

Example 13

A series RLC circuit has an impedance of 40 ohm at a frequency of 200 rad/sec. When the circuit excited by a 10V source of variable frequency the circuit resonates at a frequency of 250 rad/sec. The current at resonance is 0.5 ohm and the quality factor at resonance is 10. Determine the circuit parameters.

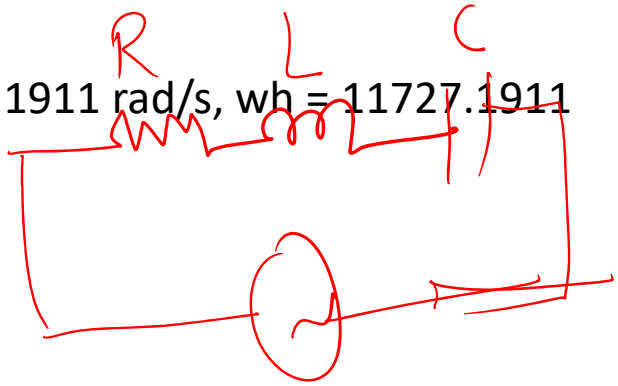
R = 20 ohm, L = 0.8 H, C = 40 uF

Example 14

A RLC series circuit consists of R = 16 ohm, L = 5 mH, & C = 2 uF. Calculate the quality factor, bandwidth and half power frequencies.

w = 10,000 rad /s, Qr = 3.125, β = 3200 rad/sec, wl = 8527.1911 rad/s, wh = 11727.1911 rad/s

$R = 16\,\Omega,\ L = 5\,mH,\ C = 2\,\mu F.$



$$Q = \frac{\omega_r L}{R} = \frac{1}{\omega_r RC}$$

$$\omega_r = \frac{1}{\sqrt{LC}} = 10{,}000 \text{ rad/sec}$$

$$Q = 3.125$$

$$\beta = \frac{\omega_r}{Q_r} = \frac{10{,}000}{3.125} = 3200 \text{ rad/sec}$$

$$\beta \text{ in Hz} = \frac{3200}{2\pi} = 509 \text{ Hz.}$$

$$\omega_l = 8527 \text{ rad/sec}$$

$$\omega_h = 11727 \text{ rad/sec}$$

$$\omega_l = \omega_r - \frac{\beta}{2} = 8400 \text{ rad/sec}$$

$$\omega_h = \omega_r + \frac{\beta}{2} = 11600 \text{ rad/sec}$$

Series RLC ckt

$$\omega_l = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_h = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$Q_r = \frac{\omega_r L}{R} \Rightarrow \boxed{\frac{R}{L} = \frac{\omega_r}{Q_r}}$$

$$\boxed{\frac{1}{LC} = \omega_r^2}$$

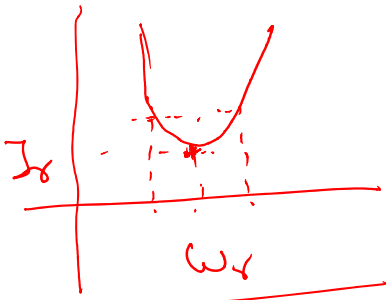
$$\omega_l = \omega_r\left(-\frac{1}{2Q_r} + \sqrt{1 + \frac{1}{4Q_r^2}}\right)$$

$$\omega_h = \omega_r\left(\frac{1}{2Q_r} + \sqrt{1 + \frac{1}{4Q_r^2}}\right)$$

Parallel RLC Ckt.

$$P_r = I_r^2 R$$

$$P = 2P_r$$



I_r at cut off freq.

$$I_{cut\ off} = \sqrt{2}\, I_r$$

$$\boxed{\frac{1}{RC} = \frac{\omega_r}{Q_r} \qquad \omega_r^2 = \frac{1}{LC}}$$

$$\omega_l = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$\omega_h = +\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$B = \omega_h - \omega_l = \frac{1}{RC}$$

$$\omega_l = \omega_r\left(-\frac{1}{2Q_r} + \sqrt{1 + \frac{1}{4Q_r^2}}\right)$$

$$\omega_h = \omega_r\left(\frac{1}{2Q_r} + \sqrt{1 + \frac{1}{4Q_r^2}}\right)$$

Example 15

The parameters of a RLC parallel circuit excited by a current source are R = 40 ohm, L = 2 mH & C = 3 uF. Determine the resonant frequency, quality factor, bandwidth and cut-off frequencies.

fr = 2.0547 kHz, Q = 1.5492, β = 8333 rad/s, Bandwidth = 1.3263 kHz, fh = 2.8222 kHz, fl = 1.4959 kHz

$$\omega_r = \frac{1}{\sqrt{LC}} = 12910 \text{ rad/sec}$$

$$f_r = \frac{12910}{2\pi} = 2.05 \text{ kHz}$$

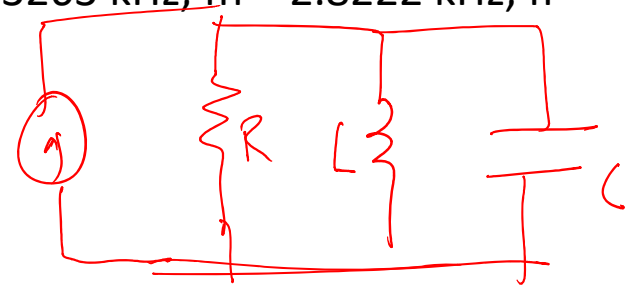
$$Q_r = \frac{R}{\omega_r L} = 1.54$$

$$\text{Bandwidth} = \beta = \frac{1}{RC} = 8333 \text{ rad/sec}$$

$$\beta \text{ in Hz} = \frac{8333}{2\pi} = 1.32 \text{ kHz}$$

$$f_l = 1.49 \text{ kHz}$$

$$f_h = 2.82 \text{ kHz}.$$



Parallel ckt.

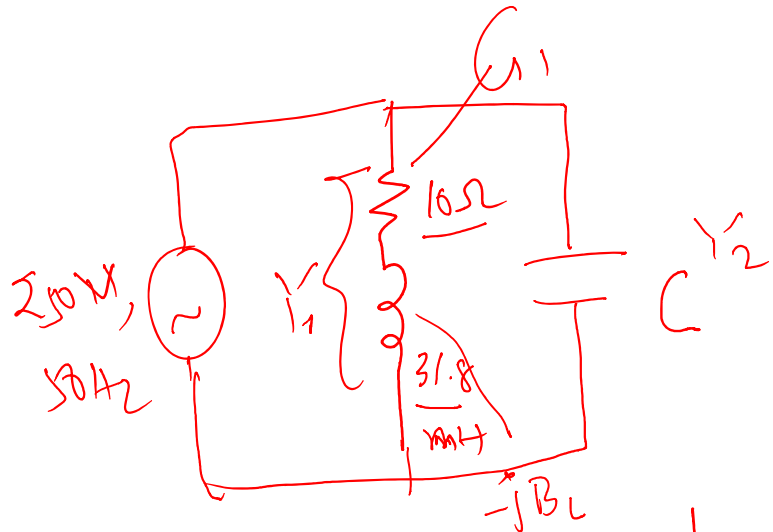
$$Q_r = \frac{R}{\omega_r L} = \omega_r RC$$

$$\beta = \frac{1}{RC} \text{ rad/sec}$$

Example 16

A coil of inductance 31.8 mH and resistance of 10 ohm is connected in parallel with a capacitor across a 250 V, 50 Hz supply. Determine the value of capacitance if no reactive current is taken from the supply.

C = 159.15 uF



$$Y = Y_1 + Y_2$$

$$= \frac{1}{R_1 + j\omega L} + jB_c$$

$$= 0.05 + j(B_c - 0.05)$$

at Resonant freq.

$$B_c - 0.05 = 0 \Rightarrow B_c = 0.05$$

$$C = 159.1\ \mu F = \frac{B_c}{2\pi f} = \frac{0.05}{2\pi \times 50}$$

$$Y_1 = \frac{1}{Z_1} = \frac{1}{R + jX_L}$$

$$Y_2 = \frac{1}{-jX_c} = jB_c$$

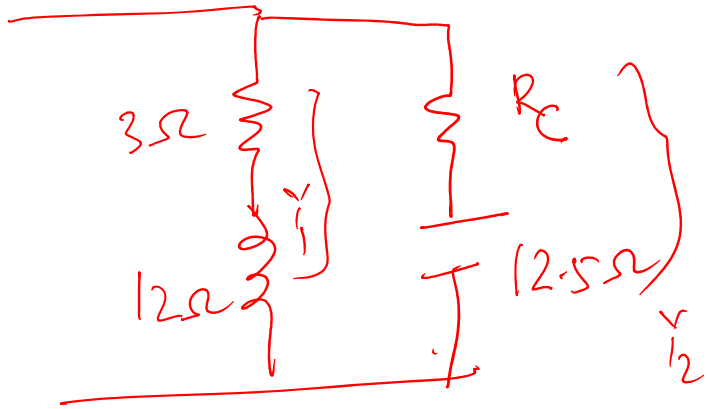
Example 17

In the RLC circuit, determine the value of R_C for resonance, If R, L is in parallel with R_C, C. Where, R = 3 ohm, X_L = 12 ohm, X_C = 12.5 ohm.

R_C = 1.7857 ohm

$R_C = ?$

$R_C = 1.7857\,\Omega$



$$Y = Y_1 + Y_2$$

$$= \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$= \frac{1}{3+j12} + \frac{1}{R_C - j12.5}$$

Separate Real & Img.

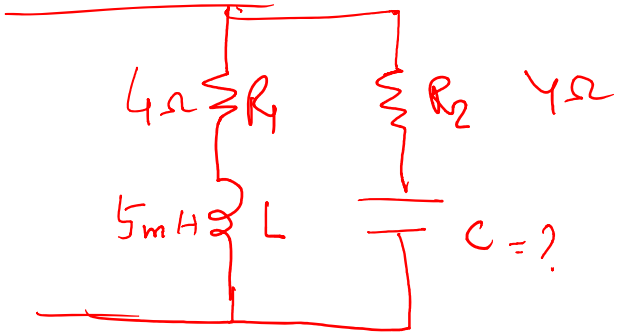
Img of $Y = 0$ $\quad R_C = 1.7857\,\Omega$

Example 18

In the RLC circuit, determine the two possible values of C for the n/w (R_1, L is in parallel with R_2, C) to resonate at 2000 rad/s. resonance. Also determine the value of C for resonance at all frequencies. R_1 = 4 ohm, L = 5 mH, R_2 = 4 ohm.

C = 49.995 uF, 312.52 uF, 312.5 uF

Resonant freq = 2000 rad/sec



$$Y = Y_1 + Y_2$$

$$= \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$\mathrm{Im}(Y) = 0$$

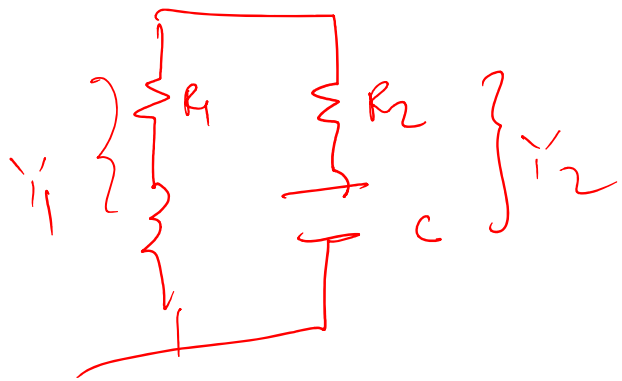
C = 49.9 MF, 312.52 MF

Example 19

Network

In the RLC circuit, determine the two possible values of L for the n/w (R_1, L is in parallel with R_2, C) to resonate at 4000 rad/s. resonance. Also determine the value of L for resonance at all frequencies. R_1 = 2 ohm, C = 20 uF, R_2 = 4 ohm.

L = 3.3693 mH, 74.2 uH



$L = ?$

$Y = Y_1 + Y_2$

$= \frac{1}{Z_1} + \frac{1}{Z_2}$ ✓ = Real + Img.

$Img(Y) = 0$

$L = 3.36$ mH

$= 74.2$ MH.